

Minimum Viable Product

LGP-16 – Kwik Report



Kwik Report developed a system that comprises a mobile app, for users to submit videos and a backend system that analyzes the videos' contents and generates metadata about them and prepares them to be sent to EVS's VIA-MAP API.

Features

Mobile app

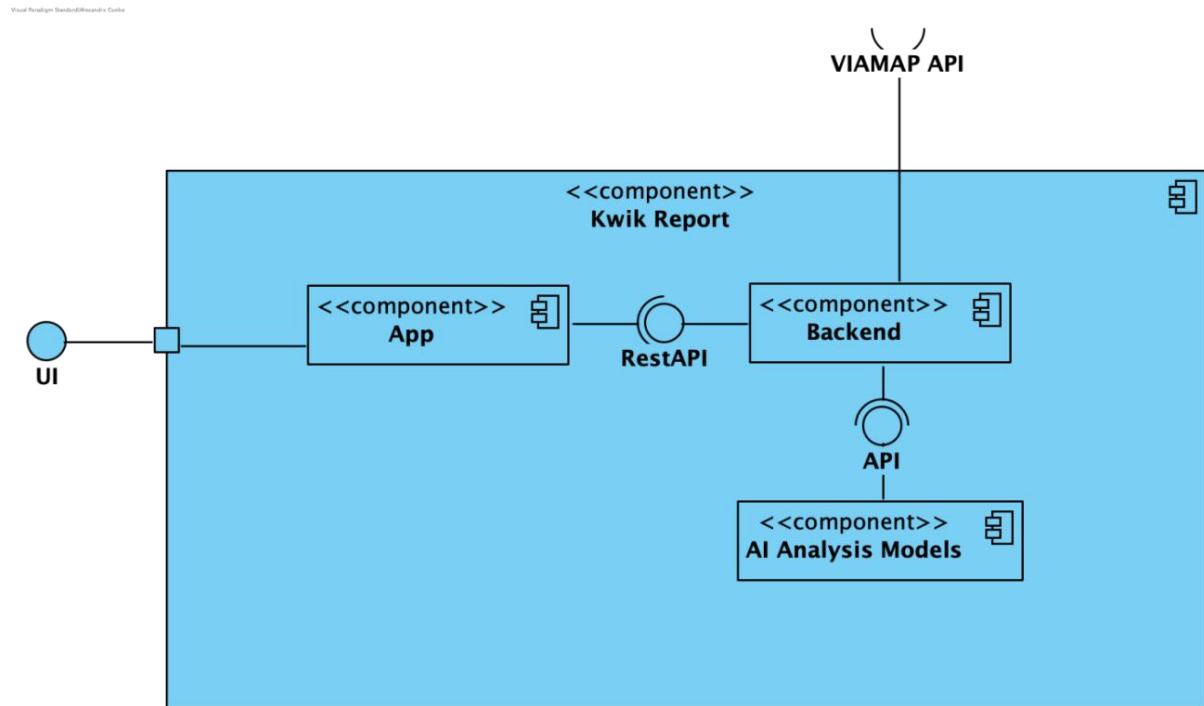
- User registration and login
- Video selection from device's gallery
- In-app video recording
- Adjustment of video location and date
- Addition of notes
- Submission to Backend

Backend Server

- Metadata extraction:
 - Video context and featured activities
 - Video quality score
 - Speech transcription and subsequent content summary
- Handling of user accounts and authentication
- Prepared to upload videos and enhanced metadata reports to VIA-MAP server

Architecture

The diagram below depicts a general view of the system's architecture.



Backend (& Analysis Models)

The component is intended to run on a server to which the Mobile App will send the video upload requests. It relies on a Docker Compose stack that ensures integration between its sub-components.

Nginx proxy

Entry point for the service, making it have a single endpoint. Protects the tusd component, acting as the authorization verifier for those who want to access it.

Rust app

Central component that sets up the application server and its routes via an HTTP API.

tusd

Implements the tus protocol, a protocol based on HTTP for resumable file uploads.

Redis

Stores authentication tokens and acts as a video queue for the workers that will analyze them.

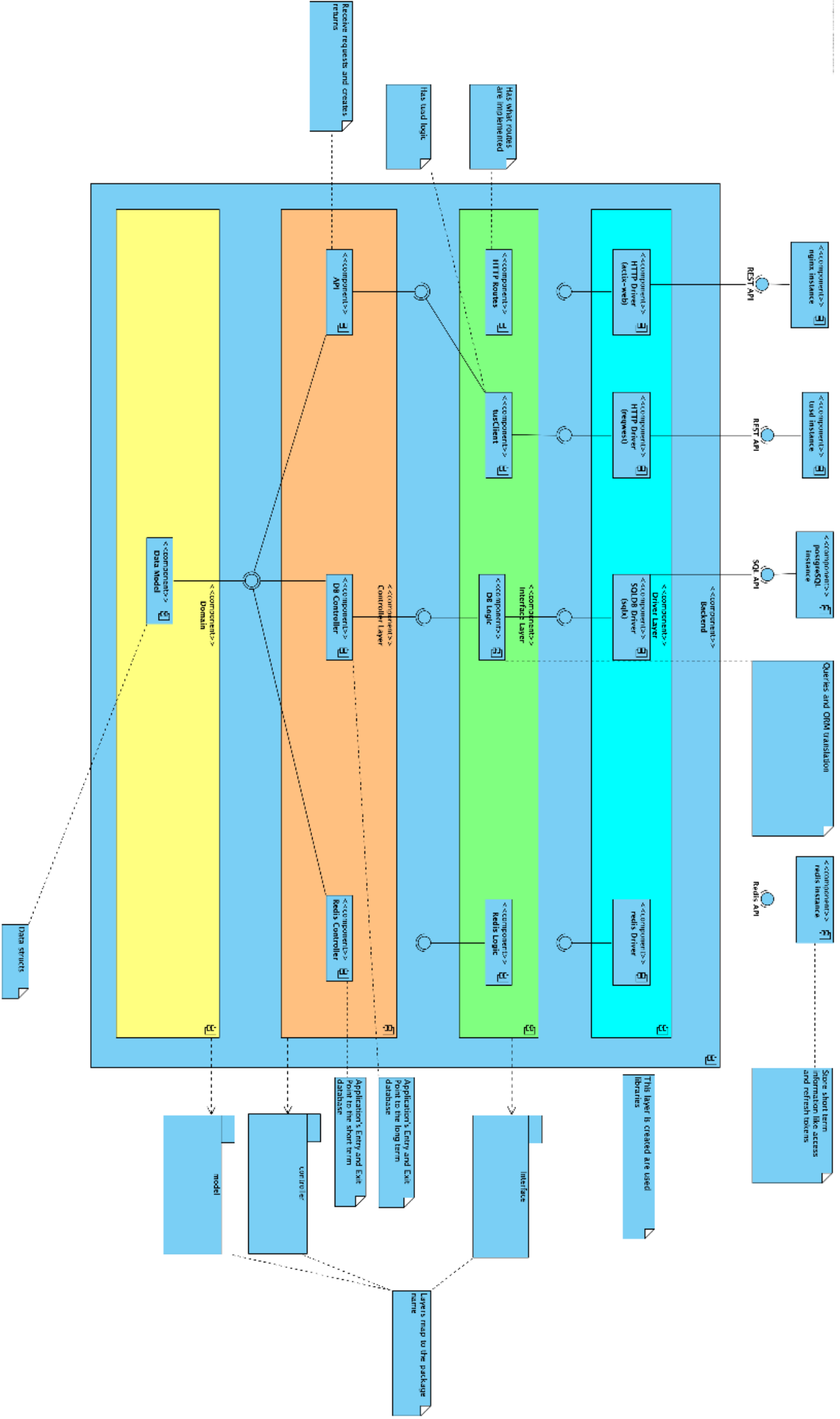
PostgreSQL database & pgAdmin

Stores user account information. pgAdmin offers an interface to interact with the database, which may come in handy since there is no direct user account deletion or edition through the mobile app or the HTTP API.

Analysis workers

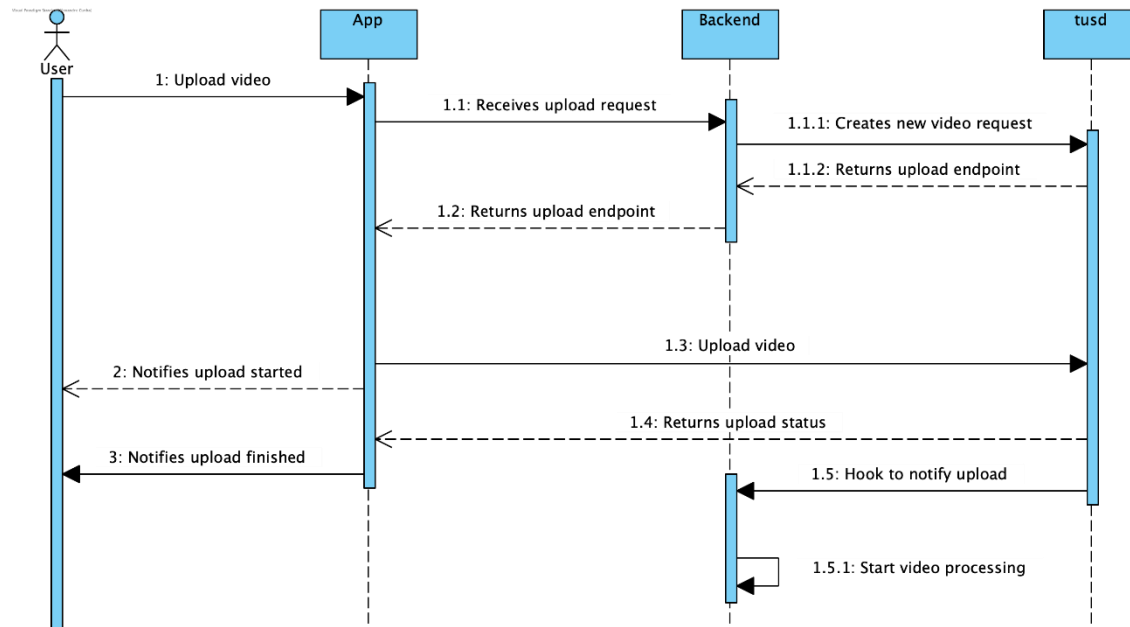
One python app for each of the three metadata generation aspects (context/activities, quality score, transcription/summary).

The diagram in the following page provides a detailed view of the described components and how they interact with each other.



Sequence Diagram

The sequence diagram below depicts the set of operations and communications occurring between the Mobile App and the Backend.



Handover process

As of June 2nd 2025, the handover process is still being defined with EVS.

Future work

A set of new features that could be developed in the future is presented below.

It is worth noting that, in the final phase of the development of the MVP, interviews were conducted by the team with staff members of the Portuguese news agencies *Público* and *JornalismoPortoNet*, where the concept behind Kwik Report was discussed. In those conversations, the staff members highlighted some features that would be of interest. Those features are highlighted with an asterisk (*).

- Detection of AI-generated content in the videos
- Identity recognition of people feature in the videos (*)
- In-app video editor (*)
- Multi-camera support